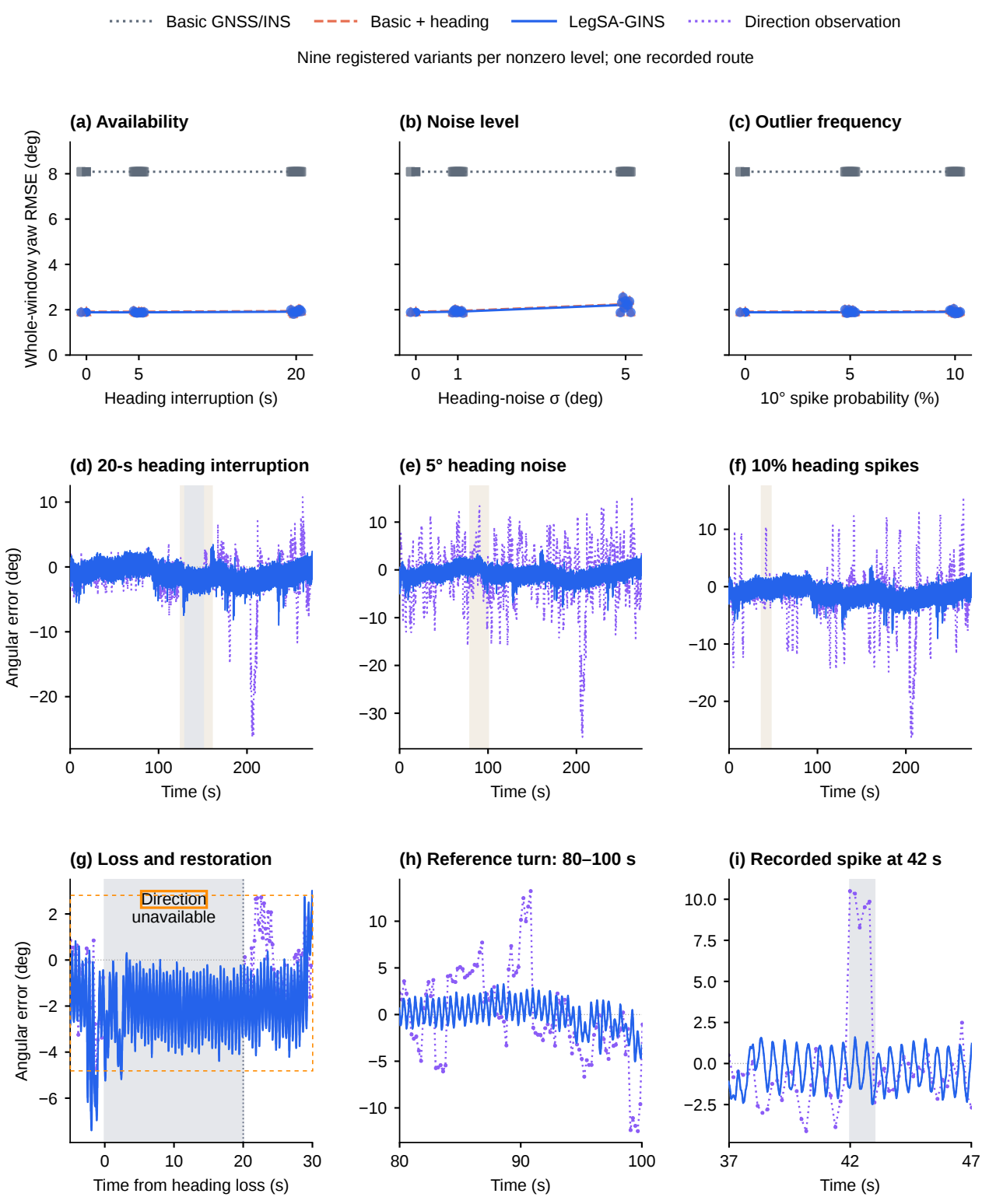

Basic GNSS/INS
Basic + heading
LegSA-GINS
Direction observation
Nine registered variants per nonzero level; one recorded route
(a) Availability
(b) Noise level
(c) Outlier frequency
Whole-window yaw RMSE (deg)
Heading interruption (s)
Heading-noise σ (deg)
10° spike probability (%)
(d) 20-s heading interruption
(e) 5° heading noise
(f) 10% heading spikes
Angular error (deg)
Time (s)
(g) Loss and restoration
(h) Reference turn: 80–100 s
(i) Recorded spike at 42 s
Direction unavailable
Time from heading loss (s)